Keyed Long-Lasting Smart Gateway [KSLG]: Implementing Mahalanobis Distance in Sensor Fusion Logic for an IoT-Based Doorbell

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***Abstract*—Current smart doorbells often suffer from high false positive rates leading to alert fatigue. KLSG, is an IoT-based smart doorbell that uses multi-sensor fusion along with the Mahalanobis distance equation to improve anomaly detection. By combining data from a camera, PIR sensor, and an accelerometer, KLSG distinguishes normal environmental activity from potential security threats with precise distinction. Results showed that a 3-hour calibration established the most stable baseline for both daytime and nighttime conditions. In comparison with Ring Doorbell Pro, KLSG achieved a 100% True Positive Rate in tampering detection while maintaining a near-zero False Positive Rate for normal pedestrian activity. Its modular hardware, real-time dashboard, and open-source framework support a reliable and sustainable residential security system.**

Keywords: Anomaly detection, Sensor fusion, Mahalanobis distance, Alert fatigue

I. Introduction

The smart doorbell market has experienced rapid growth due to the increasing integration of Internet of Things (IoT) technologies into everyday life. These devices extend traditional doorbells by enabling video monitoring, motion detection, and remote communication. The global market, valued at USD 3,480.1 million in 2022 and projected to reach USD 33,428.5 million by 2030, highlights their growing importance in modern home security systems [1]. This expansion is driven by the demand for automation and seamless integration with IoT-enabled systems such as smart locks, lighting, and home assistants [2].

With the rise of IoT, residential systems increasingly rely on multiple sensors and sensor fusion techniques to support intelligent decision-making, aiming to improve accuracy and suppress noise through multi-source data integration [3]. However, despite these advancements, current smart doorbells face critical reliability issues. Environmental variations, sensor noise, and system uncertainties often result in frequent false positives, where non-threatening events such as falling leaves, passing animals, or lighting changes trigger unnecessary alerts. This behavior is consistent with anomaly-based detection systems in IoT environments, which are known to suffer from false positive rates due to variability in real-world conditions [4]. The accumulation of these alerts leads to alert fatigue, causing users to become desensitized and potentially ignore critical security notifications.

At the same time, missed detections remain a major concern, as systems that rely primarily on vision-based motion detection may fail to identify actual tampering or intrusion events [5]. Furthermore, most commercial smart doorbells lack robust mechanisms for detecting physical tampering, such as camera obstruction or device manipulation that limits its effectiveness in real-world conditions [6].

To address these challenges, this study proposes KLSG (Keyed Long-lasting Smart Gateway), an IoT-based smart doorbell that employs multi-sensor fusion and the Mahalanobis distance metric for anomaly detection. Unlike traditional systems, this approach accounts for both variance and correlation between sensor readings; through this process, the system improves the differentiation between normal activity and potential tampering. Through this approach, KLSG aims to reduce false positive rates, improve detection accuracy, and mitigate alert fatigue while maintaining sustainable system design.

II. KLSG

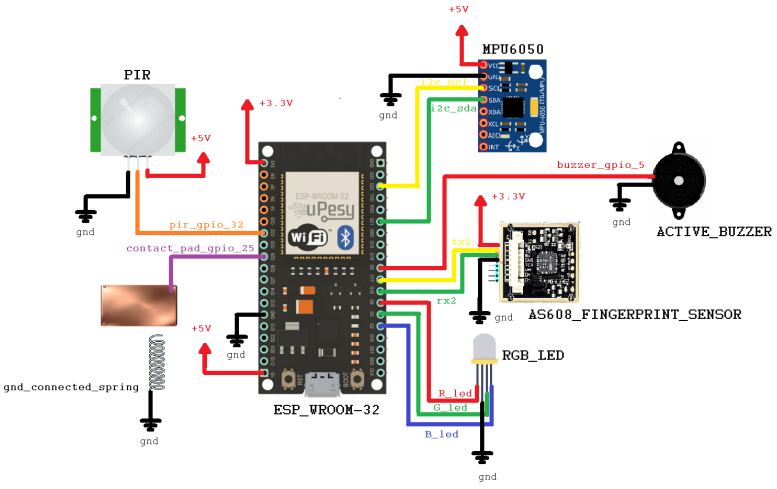


Fig. 1. KLSG Hardware Schematic

A. System Overview

KLSG is an IoT-based smart doorbell system designed to improve reliability of anomaly detection through the use of multi-sensor fusion. The system is structured to capture both

environmental activity and direct physical interaction with the device, allowing for assessment of potential security events. The system operates by continuously collecting data from multiple sensors and representing the system's state as a single value. This data is then evaluated against a baseline model derived from normal operating conditions. Deviations from this baseline are quantified using statistical anomaly detection method, enabling the system to determine whether an observed event corresponds to normal activity or a potential tampering attempt.

B. Hardware Architecture and Sensing Layer

The hardware implementation of KLSG is centered on the ESP32 microcontroller, which serves as the main processing and communication unit. The ESP32 manages sensor data aquisition, executes the anomaly detection alogirithm, and facilitates communication with external interfaces [7]. KLSG features a modular design, allowing individual hardware components to be easily replaced if they fail. The live dashboard simplifies maintenance by providing the real-time sensor readings to identify malfunctioning parts. Furthermore, the project is fully open-source, ensuring long-term software sustainability.

The sensing layer consists of multiple components:

1) **Fingerprint Module**: Used for biometric authentication of registered users and operates independently from the anomaly detection subsystem.
2) **Case Tamper Contact Pad**: Integrated into the casing to detect case prying, the mechanism utilizes a copper pad on one section of the enclosure and a spring contact on the other.
3) **Accelerometer**: Measures vibration and sudden movements.
4) **Passive Infrared Sensor (PIR)**: Detects motion by measuring changes in the infrared radiation emitted by surrounding objects, such as people or animals.
5) **Camera Module**: Provides visual input and detect anomalies by analyzing changes in pixel data.

C. Sensor Fusion and Anomaly Detection

To improve detection robustness, KLSG employs a sensor fusion framework that combines data from multiple sensors into a unified representation. Each sensor contributes a normalized measurement, forming a feature vector:

$$x = [abc] \tag{1}$$

where a represents total pixel deviation, b represents PIR output, and c represents acceleration magnitude. A baseline model is established during system calibration, where data is collected under normal operating conditions. The mean vector (μ) and covariance matrix (Σ) are computed to characterize the normal environment's behavior. Anomaly detection is performed using Mahalanobis distance, defined as:

$$D^2 = (x - \mu)^T \Sigma^{-1} (x - \mu) \tag{2}$$

This metric accounts for both variance of individual sensor measurements and the correlation between them, providing a more reliable measure of deviation compared to conventional distance metrics such as the Euclidean metric [8]. As part of the preliminary work, a comparative analysis between Euclidean and Mahalanobis was conducted to determine which method exhibits a lower false positive rate. Subjecting both distance metrics to normal operating events and deliberate tampering events. Normal events involve a person walking across the KLSG at distances of 1m, 3m, and 5m, while tampering events consisted of the person walking at those same distances while simultaneously shaking the system, triggering all of the sensors. Three detection thresholds were established using Chi-square (x^2) distributions with 3 degrees of freedom at significance levels of 0.25, 0.05 and 0.01. The preliminary results shown in Figure 2 below demonstrated that the Mahalanobis distance yielded a significantly lower false positive rate and maintained higher consistency in detecting tampering events across all distances compared to the Euclidean distance, which failed to account for the inherent correlations between the fused sensor data.

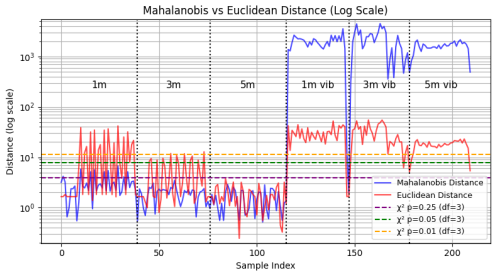


Fig. 2. Mahalanobis VS Euclidean

The three thresholds were defined at 99.75%, 99.95%, and 99.99% confidence level to classify anomaly severity. To provide immediate system feedback, an RGB LED indicator is integrated into KLSG and these color-coded alerts are mirrored directly on the live dashboard view through dynamic, color changing text (Figures 17-19). The output color is determined based on the value of D^2 relative to the defined thresholds. If D^2 exceeds the 99.75% threshold, the system outputs green, indicating early-stage anomalies. If it exceeds the 99.95% threshold, the system outputs yellow, indicating moderate-risk or suspicious activity. If it exceeds the 99.99% threshold, the system outputs red, indicating a critical anomaly associated with tampering.

D. Decision Logic and System Operation

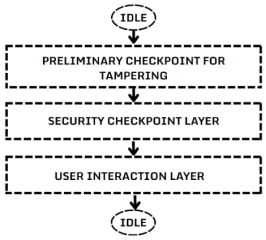


Fig. 3. Operational Flowchart

The system transitions from an Idle state into a multi-layered decision process to ensure hardware integrity and data accuracy. As illustrated in the system flowchart, the process begins at the Preliminary Checkpoint for Tampering, which monitors the contact between a copper pad and spring. The system casing is required to remain intact during normal operation; if the enclosure is opened or physically compromised, the tamper contact is triggered, immediately indicating that the system has been tampered with. In this case, an alarm is generated and the system is flagged as compromised. If no tampering is detected, the process moves to the security Checkpoint Layer. Here, the system proceeds to evaluate sensor data using fusion-based anomaly detection model. The computed distance is compared against a predefined threshold (T_d). A threshold is applied such that events with $D^2 \geq T_d$ are classified as anomalies. T_d is determined using high-confidence chi-square distribution to minimize false positives [9] [10]. Once the system confirms it is not compromised, it enters the User Interaction Layer. In this stage, KLSG functions as a smart doorbell; a fingerprint sensor is utilized for user authentication, and an alert is dispatched to the owner to facilitate interaction. After the interaction is complete, the system returns to its Idle state.

E. Monitoring and Management Interface

The KLSG system features a centralized web-based dashboard that provides the administrator with real-time oversight and administrative control. Key functionalities of this interface include:

- **Real-time monitoring**: A live video stream from the camera module allows the owner to monitor the premises remotely. As shown in Figure 4, the dashboard provides a low-latency feed for immediate visual verification. It also integrates real-time outputs from the accelerometer, PIR sensor, and pixel deviation, supplemented by continuous Mahalanobis distance computation.

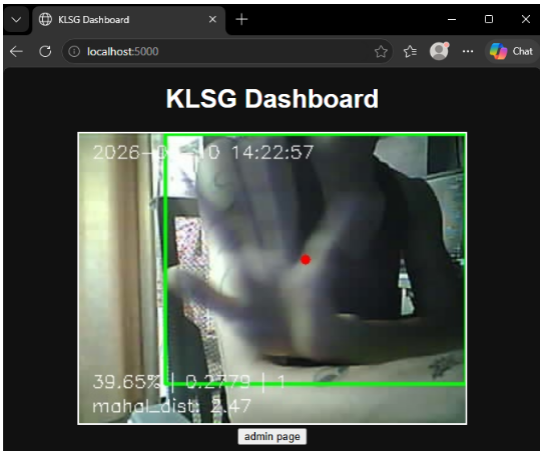


Fig. 4. KLSG Dashboard

- **User Management**: The interface provides a directory of currently registered users, including the email addresses, as illustrated in Figure 5. Administrators can manage these profiles or initiate the enrollment process for new users directly through the portal.

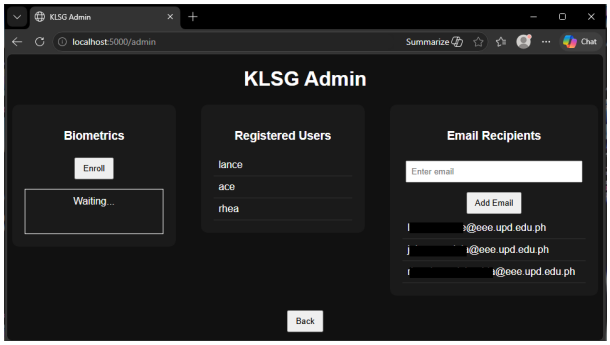


Fig. 5. User Management

- **Biometric and Anomaly Notifications**: Upon interaction with the fingerprint sensor, the system cross-references the data with the local database to determine the individual's identity. The system then dispatches an automated email notification to the owner, which explicitly alerts them whether the person is a known registered user (identifying them by name) or an unknown individual. These alerts contain a timestamped log of the event, providing the owner with immediate situational awareness. Additional real-time notifications are triggered if the system detects any security anomalies or irregularities.



Fig. 6. KLSG Notifications

F. Validation and Testing

1) Calibration and Baseline Analysis: To determine the optimal window for characterizing environmental noise, the project evaluates three distinct iterations of the baseline model. The mean vector (μ) and covariance matrix (Σ) are computed over three experimental durations: 1 hour, 2 hours, and 3 hours. Additionally, these iterations are performed during both daytime and nighttime periods to account for fluctuations in ambient light. This iterative approach aims to identify the calibration length that yields the lowest False Positive Rate (FPR) by better defining the normal environmental state and analyzing how the system's sensitivity shifts across different lighting conditions.

2) Experimental Scenarios: For each baseline iteration, the following scenarios were tested:

- Normal Events:
  1) Distance Testing: A person walking at fixed distances of 1, 3, and 5 meters away from system.
  2) Directional: A person walking directly toward and away from the system to evaluate depth-sensing response.
- Tampering and Interference Events:
  1) Forced Interaction: All proximity and directional tests (1m, 3m, 5m, toward,) were repeated while simultaneously shaking the system.
  2) Visual Obstruction: Complete covering of the camera lens to simulate masking or vandalism with simultaneous shaking of the system.
  3) Case prying: Activation of tamper contact.

3) Commercial Benchmarking: The KLSG system was benchmarked against a Ring doorbell in six experimental trials. The benchmark compares the three KLSG baseline iterations against the Ring system in two configurations:

- Standard Mode: Basic features of Ring Doorbell
- Pro Subscription: Advanced features of Ring Doorbell

4) Evaluation Metrics: Performance is quantified using True Positive Rate (TPR) for tampering detection and False Positive Rate (FPR) for normal events. Moreover, the results analyze how increasing baseline calibration time affets the system's ability to distinguish normal events from tampering events.

III. Results and Discussion

This presents and interprets the calibration results, specifically identifying the optimal baseline recording duration required to minimize false positive rates. Additionally, it provides a comparative analysis of the detection performance between the KLSG and the benchmark doorbell.

A. Calibration Results

The calibration phase was conducted to establish the baseline operating behavior of the KLSG under normal environmental conditions. Sensor data from the accelerometer, PIR, and camera module were collected during both daytime and nighttime operation without intentional tampering events. Separate calibration datasets were generated using 1-hour, 2-hour, and 3-hour collection periods to evaluate the effect of calibration duration on the stability of the baseline model. For each data set, the mean vector (μ) and covariance matrix (Σ) were computed. μ represents the average sensor response during normal operation, while Σ matrix describes the variability and correlations among sensor measurements. The covariance matrix is represented as:

$$\Sigma = \begin{bmatrix} var(a) & cov(a,b) & cov(a,c) \\ cov(a,b) & var(b) & cov(b,c) \\ cov(a,c) & cov(b,c) & var(c) \end{bmatrix}$$

wherein, a: camera output, b: PIR and c: accelerometer. Variance describes how much an individual sensor fluctuates from its mean value, while covariance describes how two sensors vary together under normal environmental conditions [11]. The computed mean vectors are presented in Table I. While, the covariance matrices are summarized in Table II.

TABLE I
Mean Vector

Calibration	Environment	Mean Vector (μ) [a b c]
1 Hour	Daytime	[3.712 0.341 0.262]
1 Hour	Nighttime	[0.880 0.242 0.306]
2 Hours	Daytime	[2.913 0.247 0.372]
2 Hours	Nighttime	[0.210 0.240 0.072]
3 Hours	Daytime	[1.731 0.254 0.282]
3 Hours	Nighttime	[1.619 0.241 0.212]

The computed mean vectors shown in Table I represent the average baseline response of a, b, and c. This mean vector serves as the reference point of normal system behavior, where incoming sensor measurements are compared using the Mahalanobis distance metric to determine significant deviations from normal conditions.

For the 1-hour and 2-hour calibration periods, daytime conditions produced higher camera mean values due to increase environmental activity and lighting variations, while nighttime conditions resulted in lower values. Meanwhile, PIR and accelerometer means remained relatively small and consistent across all calibration durations, indicating minimal fluctuations under normal circumstances. At the 3-hour calibration period, the mean vectors became more stable between daytime and nighttime operation, suggesting that longer calibration durations produce a more reliable baseline.

TABLE II
Covariance Matrix

Calibration	Environment	Covariance Matrix (Σ)
1 Hour	Daytime	$\begin{bmatrix} 4.176 & -0.007 & 1.199 \\ -0.007 & 0.157 & 0.006 \\ 1.199 & 0.006 & 0.193 \end{bmatrix}$
1 Hour	Nighttime	$\begin{bmatrix} 6.021 & -0.002 & 0.205 \\ -0.002 & 0.009 & 0.005e-02 \\ 0.205 & 0.005e-02 & 0.213 \end{bmatrix}$
2 Hours	Daytime	$\begin{bmatrix} 49.225 & 0.006 & 0.750 \\ 0.006 & 0.010 & 0.002 \\ 0.750 & 0.002 & 0.234 \end{bmatrix}$
2 Hours	Nighttime	$\begin{bmatrix} 2.358 & 0.004e-01 & 0.125 \\ 0.004e-01 & 0.001e-01 & -0.003e-02 \\ 0.125 & -0.003e-02 & 0.067 \end{bmatrix}$
3 Hours	Daytime	$\begin{bmatrix} 13.611 & -0.001 & 0.440 \\ -0.001 & 0.004 & 0.001 \\ 0.440 & 0.001 & 0.202 \end{bmatrix}$
3 Hours	Nighttime	$\begin{bmatrix} 0.157 & -0.006e-02 & 0.620 \\ -0.006e-02 & 0.001e-01 & 0.001e-02 \\ 0.620 & 0.001e-02 & 0.167 \end{bmatrix}$

The computed results in Table II showed noticeable differences between daytime and nighttime conditions. In the 1-hour and 2-hour datasets, Daytime conditions exhibited significantly higher variance peaking at 49.225 likely due to increased ambient noise and activity. In contrast, Nighttime conditions generally produced much lower covariance magnitudes, such as 2.358 recorded at 2 hours, suggesting a more stable baseline. By the 3-hour mark, the system reached its most consistent state; while the daytime variance remained elevated at 13.611, the nighttime variance dropped to a minimum of 0.157. These indicate that longer calibration periods, particularly during

nighttime, allow the system to better supress short-term noise and characterize a more stable environmental baseline.

B. Detection Performance

The detection performance of the KLSG was evaluated using both tampering and non-tampering scenarios to determine the effectiveness of the sensor fusion in distinguishing abnormal intrusion events from normal environmental activity. The experiments were conducted using mutiple controlled trials involving physical tampering events and ordinary environmental disturbances. Tampering scenarios include enclosure opening and device shaking, while non-tampering scenarios included distant pedestrian movements, and ambient environmental motion. For each trial, KLSG recorded the sensor readings, computed Mahalanobis distance (D^2), and compared the resulting value against the established anomaly threshold. The average sensor readings and corresponding Mahalanobis distance are presented in Table III.

TABLE III
SCENARIO-BASED AVERAGE SENSOR READINGS AND D^2 UNDER DAYTIME AND NIGHHTTIME CONDITIONS

Scenario	Env.	Accel.	PIR	Cam.	Avg. D^2	Output
PM (5m)	N	0.240	0.533	2.407	2.903	NA
PM(5m)	D	0.296	0.452	3.745	2.016	NA
PM (3m)	N	0.239	0.645	5.465	4.264	NA
PM (3m)	D	0.245	0.800	11.485	2.998	NA
PM(1m)	N	0.242	0.727	11.340	8.886	NA
PM(1m)	D	0.244	0.600	18.216	5.041	NA
PM (A)	N	0.244	0.515	5.659	4.649	NA
PM (A)	D	0.262	0.485	8.272	2.605	NA
PM (T)	N	0.238	0.438	3.046	3.474	NA
PM (T)	D	0.297	0.485	5.769	2.533	NA
CO	N	0.273	0.125	0.425	3.358	NA
CO	D	0.354	0.235	11.498	4.059	NA
PM-S (5m)	N	9.6184	0.533	2.871	856.546	A
PM-S (5m)	D	9.770	0.516	10.109	150.896	A
PM-S (3m)	N	9.831	0.800	7.385	875.882	A
PM-S (3m)	D	6.651	0.710	31.529	101.594	A
PM-S (1m)	N	9.527	0.871	19.659	848.020	A
PM-S (1m)	D	8.807	0.690	29.369	135.964	A
PM-S (A)	N	12.709	0.531	7.171	1138.810	A
PM-S (A)	D	8.763	0.567	17.583	135.055	A
PM-S (T)	N	9.364	0.552	6.083	833.270	A
PM-S (T)	D	6.325	0.344	13.191	96.360	A
CO-S	N	8.960	0.242	0.506	796.475	A
CO-S	D	7.154	0	0.2	109.376	A

Note: PM: Pedestrian Movement, CO: Camera Obstruction, S: Shaking, A: Away, T: Towards D: Daytime, N:Nightime, NA: No Alert, A: Alert.

The results in Table III confirm that the sensor fusion framework effectively distinguishes between routine environmental activity and deliberate tampering using a Chi-square threshold of 11.645. Under normal event conditions, the average D^2 values consistently remained below this limit, ranging from 2.016 to 8.886 ensuring a 'No alert' status despite environmental fluctuations from pedestrian movement. While daytime activity generally produced higher raw sensor deviations, it is correctly identified as statistically consistent with normal baseline noise. In contrast, tampering scenarios triggered a massive surge in anomaly score that far exceeded the threshold, with D^2 values jumping to a range between 96.360 and 1138.810. This drastic increase was primarily driven by the accelerometer, which spiked from a baseline of 0.212 to 12.709 during physical interference. KLSG maintained a clear statistical margin between normal and tampering event, successfully triggering 'Alert' outputs across all tampering events regardless of lighting conditions.

The performance of the anomaly detection framework was further evaluated using the True Positive Rate (TPR) and False Positive Rate (FPR). The TPR measures the proportion of actual tampering events correctly identified by the system, while the FPR measures the proportion of normal events incorrectly classified as tampering events.

C. KLSG Benchmarking

The detection performance of KLSG was benchmarked against Ring Doorbell Pro operating with an active subscription service that enables advanced features such as video preview alerts, doorbell calls, customizable smart alerts, and person, package, and vehicle alerts. Preliminary calibration results showed that the 3-hour baseline configurations of the KLSG produced the most stable covariance structure and the lowest false positive behavior among all tests calibration durations. Following this optimization, detection outputs from all experimental scenarios were integrated into unified master confusion matrices. These matrices (Figures 7-8), served as the basis for comparative analysis of the TPR and FPR for both systems.

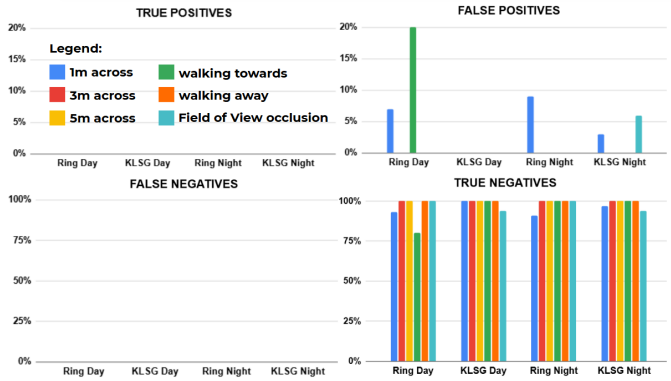


Fig. 7. Confusion Matrix (Normal Events)

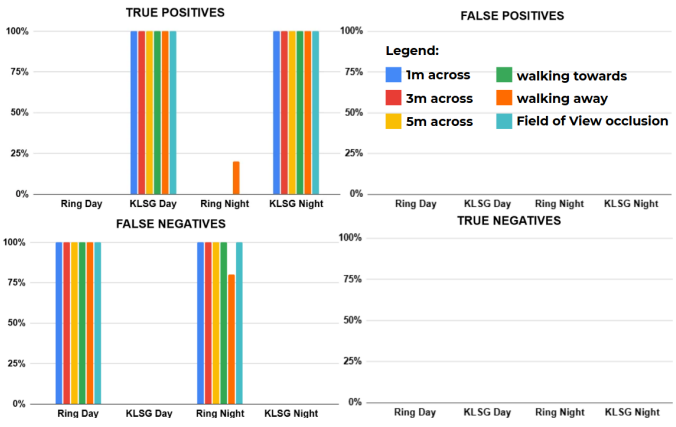


Fig. 8. Confusion Matrix (Tampering Events)

Since universal standards for intrusion detection are still evolving in intelligent systems, existing literature such as [12]

and [13] recognizes these targets as important metrics for the reliability of the system and the evaluation of the effectiveness of anomaly detection in smart systems:

- False Positive Rate (FPR): $\leq 10\%$
- False Negative Rate (FNR): 0%
- True Positive Rate (TPR): $\geq 92 - 95\%$
- True Negative Rate (TNR): $\geq 90\%$

The comparative analysis between KLSG and Ring demonstrates the superior reliability of KLSG, which achieved a consistent 100% true positive rate across all tampering scenarios while the Ring exhibited nearly 100% false negatives; this is primarily due to the absence of tampering detection, as physical shaking does not significantly affect the Ring's stabilized camera output. KLSG maintained high environmental robustness in both daytime and nighttime conditions, yielding a true negative rate near 100% and only minor false positive spikes during field-of-view obstruction or close-range movement. Conversely, the Ring also exhibited higher susceptibility to false alarms at night particularly during 'walk towards' scenarios.

IV. Conclusion

The development and evaluation of KLSG showed that integrating multi-sensor fusion with the Mahalanobis distance metric significantly enhances the reliability of IoT-based security systems. By accounting for the variance and correlation between camera pixel deviation, PIR data, and accelerometer readings, KLSG effectively addresses the challenges of alert fatigue and physical security common in commercial smart doorbells. Experimental results indicate that a 3-hour calibration duration provides the most stable baseline for characterizing environmental noise, allow KLSG to maintain high accuracy across both daytime and nighttime conditions. KLSG's performance highlights a critical advantage over commercial benchmarks; while standard devices often rely solely on vision-based motion detection, KLSG achieved a 100% TPR in detecting possible tampering. Furthermore, KLSG maintained a near-zero FPR, successfully distinguishing routine pedestrian movement from actual security threats.

V. Recommendation

Future improvements to the KLSG system may include implementing an adaptive running baseline, where old sensor data are continuously replaced with newer normal-condition data to allow the system to better adapt to changing environmental conditions and reduce false positives over time. The notification system may also be enhanced by replacing email alerts with real-time push notifications for faster and more accessible user response during tampering events. On the hardware side, future implementations may utilize a microcontroller unit with an integrated camera module to allow a single-device configuration instead of separate camera and controller setups. This may simplify system integration, reduce hardware complexity, and improve overall compactness. Additionally, future hardware revisions may focus on creating a more compact and durable enclosure with fewer gaps. Integrating a built-in battery monitoring system accessible through the webpage interface may also improve usability and system maintenance. Furthermore, conducting a user-feedback survey among various age groups to have a more varied evaluation of the system. Lastly, current project conditions point the study at a technology readiness level (TRL) of 6 (full-scale prototype and field demonstration). Continuing the improvements to foster full-scale prototyping in commercial conditions to increase the project's TRL to level 7 [14].

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APPENDIX



Fig. 9. 1-hour Calibration in Normal Daytime Conditions

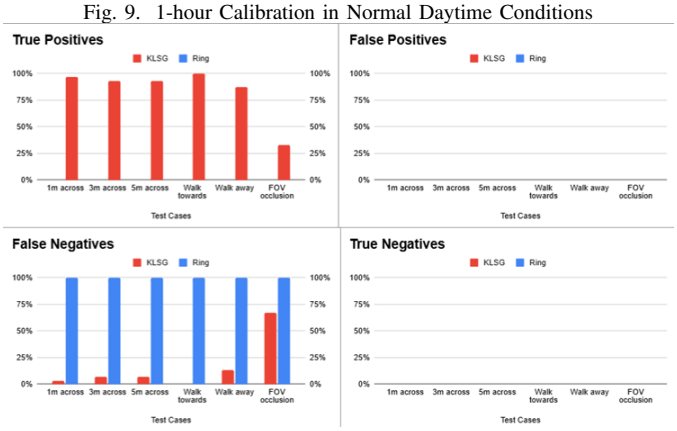


Fig. 10. 1-hour Calibration Under Daytime Tampering Conditions

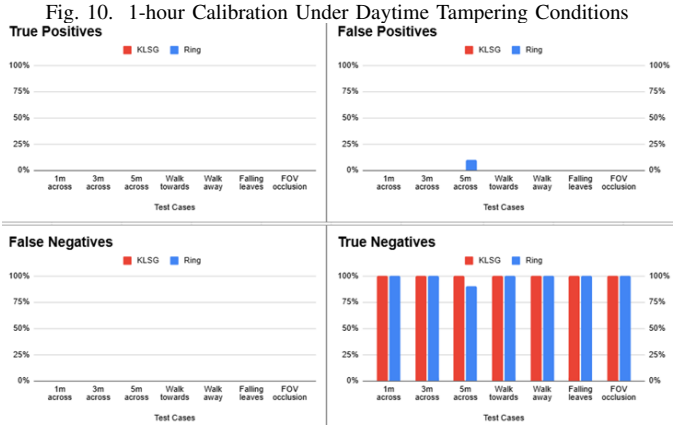


Fig. 11. 1-hour Calibration in Normal Nighttime Conditions

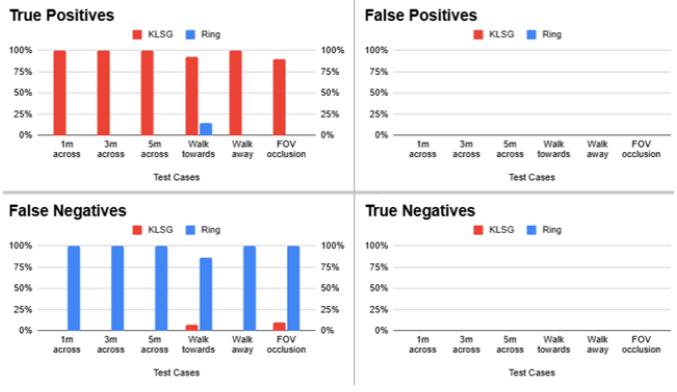


Fig. 12. 1-hour Calibration Under Nighttime Tampering Conditions



Fig. 13. 2-hour Calibration in Normal Daytime Conditions

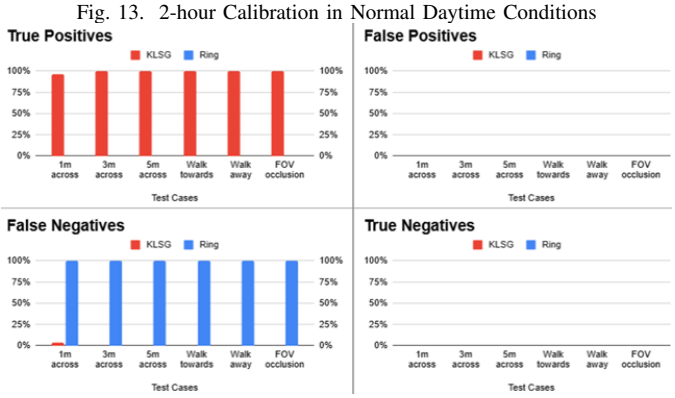


Fig. 14. 2-hour Calibration Under Daytime Tampering Conditions

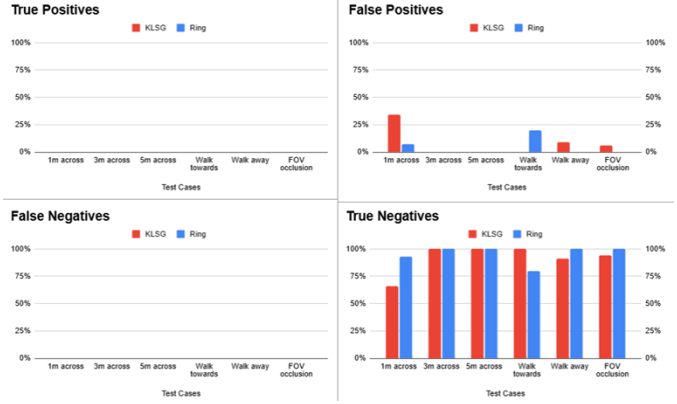


Fig. 15. 2-hour Calibration in Normal Nighttime Conditions

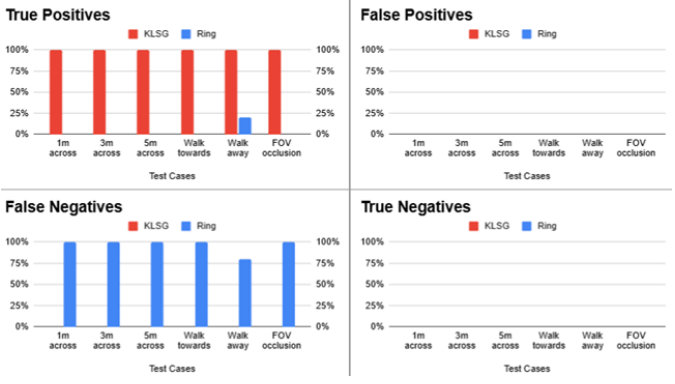


Fig. 16. 2-hour Calibration Under Nighttime Tampering Conditions

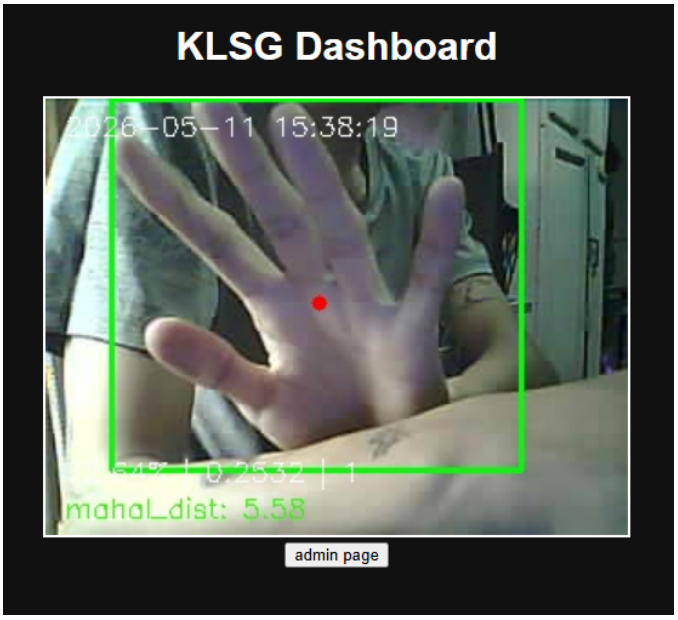


Fig. 17. Green: D^2 exceeds 99.75% threshold

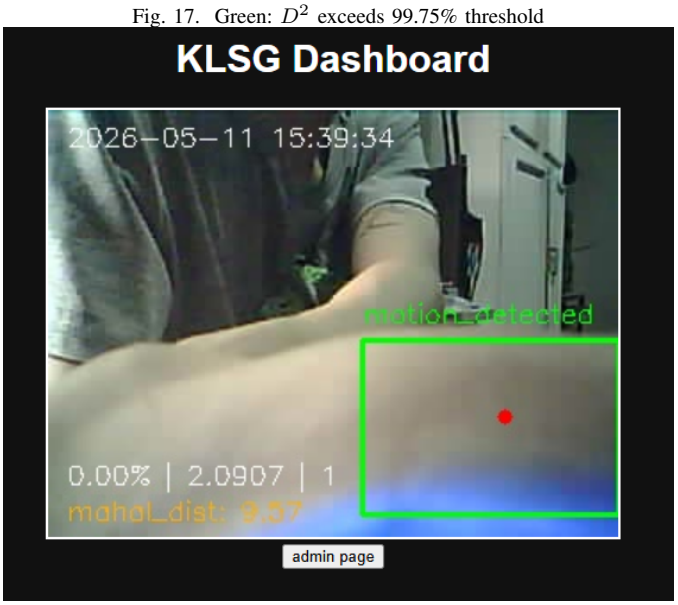


Fig. 18. Yellow: D^2 exceeds 99.95% threshold

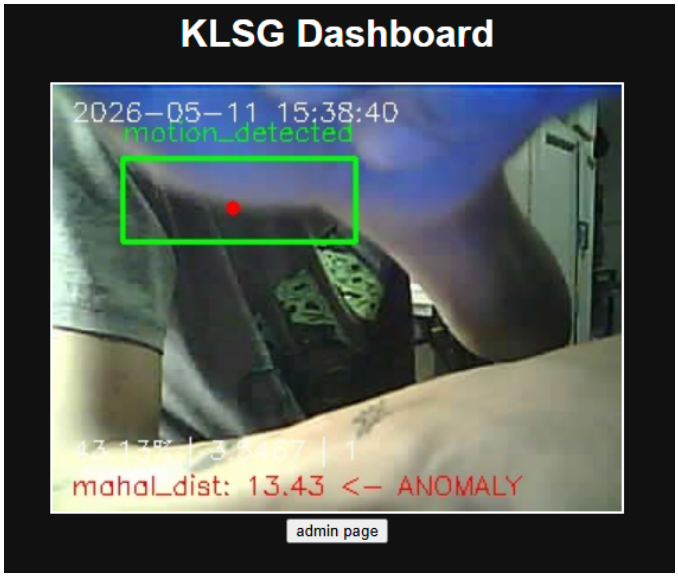


Fig. 19. Red: D^2 exceeds 99.99% threshold [Alert]



Fig. 20. Hardware implementation of the KLSG prototype



Fig. 21. Front view of the KLSG prototype during active operation